

Checkpoint Test Analysis and Design

Equivalence Partitions

Rules for the Age Field

- Must be a **whole number (integer)**
- **Minimum age:** 18 (inclusive)
- **Maximum age:** 100 (inclusive)

Equivalence Partitions

Partition ID	Type	Description	Example Values
VP1	Valid Partition	Integers between 18 and 100	18, 50, 100
IP1	Invalid Partition	Integers less than 18	0, 17
IP2	Invalid Partition	Integers greater than 100	101, 200

Test Cases Table

Test Case ID	Type	Preconditions	Steps to Execute	Expected Result	Environment	Final Result
TC-VP-001	Functional	Age field is visible and ready for input	1. Enter 50 in the Age field. 2. Click Send	System accepts the input and produces a success message: Valid age! / Âge valide !	Web browser (Chrome)	Pass
TC-IP1-001	Functional	Age field is visible and ready for input	1. Enter 17 in the Age field. 2. Click Send .	System displays error message: The age is not valid!	Web browser (Chrome)	Pass

TC-IP2-001	Functiona 1	Age field is visible and ready for input	1. Enter 101 in the Age field. 2. Click Send.	System displays error message: The age is not valid!	Web browser (Chrome)	Pass
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Summary

- 1 Valid Partition Test Case
- 2 Invalid Partition Test Cases
- Total 3 test cases

Decision Table Exercise – Password Validation

Conditions	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5
Password length ≥ 8 characters	N	Y	Y	Y	Y
Contains Uppercase character ≥ 1	-	N	Y	Y	Y
Contains 1 or more Numbers	-	-	N	Y	Y
Contain special characters	-	-	-	N	Y

Action: Accept Password	N	N	N	N	O
Action: Display Error Message	O	O	O	O	N

Decision Table - Test Case Mapping

Rule	Test Case ID	Test Input	Expected Result
Rule 1	TC-01	Pass1!@ (7 characters)	Password rejected - Password too short
Rule 2	TC-02	passwo@1 (No Uppsecase character)	Password rejected - Password does not contain an uppercase character
Rule 3	TC-03	passwo@Y (No Number / numeric character)	Password rejected - missing Number
Rule 4	TC-04	Password123 (no special character)	Password rejected - missing special character
Rule 5	TC-05	ValidPass1!	Password accepted

Test Case Execution Table

Environment: Windows 11, Chrome Browser

TC ID	Type	Preconditions	Test Data	Test Steps	Expected Result
TC-01	Functional	User is on password input page	Pass1!@ (Password < 8)	1. Enter "Pass1!@" 2. Click Send	System rejects password and shows "Password too short."
TC-02	Functional	User is on password input page	passwo@1 (No Uppercase)	1. Enter "passwo@1" 2. Click Send	System rejects password and shows "Password must contain at least 1 uppercase."
TC-03	Functional	User is on password input page	passwo@Y (No number)	1. Enter "Passwo@Y" 2. Click Send	System rejects password and shows "Password must contain at least one number."
TC-04	Functional	User is on the password input page	Password123 (No special character)	1. Enter "Password123" 2. Click Send	System rejects password and shows "Password must contain at least one special character."

TC-05	Functiona l	User is on the password input page	ValidPass1!	1. Enter "ValidPa ss1!" 2. Click Send	System accepts password
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Boundary Value Analysis (BVA) – Password Validation

Requirement

The password must contain between 8 and 20 characters (inclusive).

Testing Environment

Item	Description
Device	PC / Laptop
Operating System	Windows 11
Browser	Latest version of Google Chrome
Application URL	https://emnamagri.github.io/boite-noire/BVA.html

Summary of Boundary Value Analysis

Boundary Type	Value Tested	Purpose
Minimum - 1	7	Test invalid value just below minimum
Minimum	8	Test lowest valid value
Nominal Value	12	Test typical valid input
Maximum	20	Test highest valid value
Maximum + 1	21	Test invalid value just above maximum

Test Case Specification Matrix (BVA)

TC ID	Type	Preconditions	Steps to Execute	Test Data	Expected Result	Result (OK/KO)
TC-01	Functional	Password validation page is loaded and the input field is empty	1. Enter a password with 7 characters 2. Click "To Validate"	1234567	System displays error message: The password is not valid! It must contain between 8 and 20 characters.	OK
TC-02	Functional	Password validation page is loaded and	1. Enter a password with 8 characters 2. Click	12345678	System accepts the password (minimum valid boundary) With Success	OK

		the input field is empty	"To Validate"		Message: Valid password!/Mot de passe valide !	
TC-03	Functional	Password validation page is loaded and the input field is empty	1. Enter a password with 12 characters 2. Click "To Validate"	123456789012	System accepts the password (valid nominal value) With Success Message: Valid password!/Mot de passe valide !	OK
TC-04	Functional	Password validation page is loaded and the input field is empty	1. Enter a password with 20 characters 2. Click "To Validate"	12345678901234567890	System accepts the password (maximum valid boundary) With Success Message: Valid password!/Mot de passe valide !	OK
TC-05	Functional	Password validation page is loaded and the input field is empty	1. Enter a password with 21 characters 2. Click "To Validate"	123456789012345678901	System displays error message: The password is not valid! It must contain between 8 and 20 characters.	OK

State Transition Table Exercise - Password Validation

State Transition Matrix

From \ To	S1: Initial State	S2: Password Entered	S3: Valid Password	S4: Invalid Password
S1: Initial State	_	Valid (T1)	Invalid	Invalid
S2: Password Entered	Invalid	_	Valid (T2)	Valid (T3)
S3: Valid Password	Invalid	Invalid	_	Invalid
S4: Invalid Password	Invalid	Valid (T4)	Invalid	_

Transition Definitions

Transition	Description
T1	User enters characters in the password field
T2	Password meets validation rules (length, uppercase, number, special character)
T3	Password does not meet validation rules
T4	User edits or clears the password after an error

Transition Test Case

Environment:

- Device: PC / Laptop

- **Browser: Chrome**
- **URL:https://emnamagri.github.io/boite-noire/transition.html**

ID	Type	Preconditions	Steps	Expected Result	Status
TC_01	Functiona l	System in Initial State (S1)	1. Enter a valid password: Pass@123 2. Click Validate	System transitions from S1, S2 to S3 and displays a success message	OK
TC_02	Functiona l	System in Initial State (S1)	1. Enter an invalid password. Password 2. Click Validate	System transitions from S1, S2 to S4 and shows validation error	OK
TC_03	Functiona l	System currently in Invalid State (S4)	1. Edit password to Pass@123 2. Click Validate	System transitions from S4, S2 to S3 and shows success message	OK
TC_04	Functiona l	System in Initial State (S1	1. Leave password empty 2. Click Validate	System shows error and moves to Invalid State (S4)	OK